

Introduction to tidyverse

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1 The tidyverse Style Guide

canyoureadthissentence?

Good coding style is like correct punctuation: you can manage without it, but it makes things easier to read.

The most important thing about the tidyverse style guide is that it provides **consistency**, making code easier to write because you need to make fewer decisions.

Tip

See: [The tidyverse style guide](#)

File Names

Avoid special characters in file names — stick with numbers, letters, -, and _.

```
# Good
fit_models.R
utility_functions.R

# Bad
fit models.R
foo.r
stuff.r
lab1.r
```

Syntax

- Leave spaces before and after operators (e.g. <-, =).
- Put a space after a comma, not before.
- Object names should be all lowercase, with words separated by an underscore.

```
# Good
age <- c(6, 9, 15)
# Bad
age <- c(6,9,15)

# Good
data.frame(age_kid = age)
# Bad
data.frame(Age_kid=age)

# Good
day_one <- 1
day_1 <- 1
# Bad
DayOne <- 1
first_day_of_the_month <- 1
```

2 The Pipe Operator

What is the average of 4, 8, 16 — approximately?

Breaking the problem down:

1. What is the average of 4, 8, 16 approximately?

2. What is the **average** of 4, 8, 16 approximately?
3. What is the average of 4, 8, 16 **approximately**?

i Solution 1: Functions within Functions

```
c(4, 8, 16)
```

```
[1] 4 8 16
```

```
mean(c(4, 8, 16))
```

```
[1] 9.333333
```

```
round(mean(c(4, 8, 16)))
```

```
[1] 9
```

Problem: Things get messy and harder to read as operations become more complex.

i Solution 2: Creating Objects

```
numbers <- c(4, 8, 16)  
numbers
```

```
[1] 4 8 16
```

```
avg_number <- mean(numbers)  
avg_number
```

```
[1] 9.333333
```

```
round(avg_number)
```

```
[1] 9
```

Problem: We end up with too many objects cluttering the **Environment**.

Solution 3: The Pipe Operator |>

First, load the tidyverse:

```
# install.packages("tidyverse")  
library(tidyverse)
```

The pipe |> passes the output of one function as the first argument of the next.

Keyboard shortcut: Ctrl/Cmd + Shift + M

```
c(4, 8, 16) |>  
  mean() |>  
  round()
```

```
[1] 9
```

Read this as: Combine 4, 8, and 16, **then** take the mean, **then** round the result.

This is equivalent to the composite function

$$f \circ g \circ h(x)$$

, or round(mean(c(4, 8, 16))) — but much easier to read.

Generalized form

```
h(x) |>  
  g() |>  
  f()
```

Our example

```
c(4, 8, 16) |>  
  mean() |>  
  round()
```